

Lecture Notes: Planar Graphs and Graph Colorings

Discrete Mathematics

Author: Bakdaulet Sotsial

Astana IT University

Contents

1	Planar Graphs — Definitions and Basic Concepts	2
2	Euler’s Formula and Consequences	3
2.1	Euler’s Formula	3
2.2	Corollaries of Euler’s Formula	3
2.3	Non-Planarity of $K_{3,3}$	4
3	Kuratowski’s Theorem and Planarity Testing	4
3.1	Subdivisions and Homeomorphic Graphs	4
3.2	Graph Minors	5
3.3	Kuratowski’s and Wagner’s Theorems	5
3.4	The Petersen Graph	5
4	Graph Coloring — Vertex Coloring	6
4.1	Standard Examples	6
4.2	Bounds on the Chromatic Number	6
5	The Four Color Theorem and Planar Graph Coloring	7
5.1	Historical Context	7
5.2	The Six Color Theorem	7
5.3	The Five Color Theorem	8
6	Edge Coloring and Chromatic Index	8
6.1	Vizing’s Theorem	9
6.2	König’s Line Coloring Theorem	9

§ 1. Planar Graphs — Definitions and Basic Concepts

We begin by establishing the foundational vocabulary of planarity. Throughout these notes, a *graph* $G = (V, E)$ is a finite, simple, undirected graph unless stated otherwise.

Definition 1.1 (Planar graph). A graph G is said to be **planar** if it can be drawn in the Euclidean plane $\mathbb{R}^2$ in such a way that no two edges cross one another (except possibly at shared endpoints).

Definition 1.2 (Plane graph / planar embedding). A **plane graph** is a particular drawing (embedding) of a planar graph in the plane in which no two edges intersect except at their common endpoints. Such a drawing is called a **planar embedding** of G .

Remark 1.1. The distinction between *planar* and *plane* is important: “planar” is an intrinsic property of the abstract graph, while “plane graph” refers to a concrete drawing. A single planar graph may admit many non-equivalent plane embeddings.

Definition 1.3 (Faces of a plane graph). Let G be a plane graph. The complement $\mathbb{R}^2 \setminus G$ (the plane minus the drawing of G) is an open set whose connected components are called the **faces** (or **regions**) of G .

By the Jordan Curve Theorem, every simple closed curve in $\mathbb{R}^2$ divides the plane into exactly two components. Consequently:

- Exactly one face is **unbounded** (extends to infinity); it is called the **outer face** or **exterior face**.
- All remaining faces are **bounded** (finite regions enclosed by edges).

Definition 1.4 (Boundary walk and face degree). The **boundary walk** of a face f is the closed walk that traces the edges incident to f . The **degree** of f , denoted $\deg(f)$, is the number of edges in this boundary walk (counting each edge at most twice if it borders f on both sides).

Example 1.1 (Planar embedding of K_4). The complete graph K_4 on four vertices is planar. Figure 1 shows a planar embedding. This embedding has $v = 4$ vertices, $e = 6$ edges, and $f = 4$ faces (three bounded triangular faces f_1, f_2, f_3 and one unbounded outer face f_4). The reader may verify: $4 - 6 + 4 = 2$.

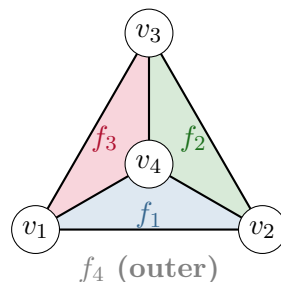


Figure 1: A planar embedding of K_4 . The three shaded regions are the bounded faces f_1, f_2, f_3 ; the exterior region is the outer face f_4 .

§ 2. Euler's Formula and Consequences

2.1. Euler's Formula

Theorem 2.1 (Euler's Formula, 1758). *Let G be a connected plane graph with v vertices, e edges, and f faces. Then*

$$v - e + f = 2.$$

Proof. We proceed by induction on the number of edges e .

Base case ($e = v - 1$). Since G is connected and has $e = v - 1$ edges, G is a spanning tree. A tree has no cycles, so its plane embedding has exactly one face (the outer face), i.e. $f = 1$. Then

$$v - e + f = v - (v - 1) + 1 = 2. \quad \checkmark$$

Inductive step. Suppose the formula holds for all connected plane graphs with fewer than e edges, where $e \geq v$. Let G be a connected plane graph with v vertices and e edges. Since $e \geq v > v - 1$, G contains at least one cycle, so there exists an edge $\ell \in E(G)$ that lies on a cycle.

Remove ℓ to obtain the plane graph $G' = G - \ell$. Since ℓ lies on a cycle, its removal does not disconnect G , so G' is still connected. Moreover, ℓ is incident to exactly two distinct faces in G ; removing it merges those two faces into one. Therefore G' has

$$v' = v, \quad e' = e - 1, \quad f' = f - 1.$$

By the inductive hypothesis applied to G' :

$$v' - e' + f' = 2 \implies v - (e - 1) + (f - 1) = 2 \implies v - e + f = 2.$$

This completes the induction. □

2.2. Corollaries of Euler's Formula

We now derive several important consequences. Throughout, G denotes a connected simple planar graph unless stated otherwise.

Corollary 2.1. *If G is a connected simple planar graph with $v \geq 3$ vertices and e edges, then*

$$e \leq 3v - 6.$$

Proof. Fix any planar embedding of G . Since G is simple and $v \geq 3$, every face has degree at least 3 (each face is bounded by at least a triangle). Summing face degrees:

$$\sum_{i=1}^f \deg(f_i) \geq 3f. \tag{1}$$

On the other hand, each edge borders exactly two faces (or the same face twice), so

$$\sum_{i=1}^f \deg(f_i) = 2e. \tag{2}$$

Combining (1) and (2): $2e \geq 3f$, i.e. $f \leq \frac{2e}{3}$. Substituting into Euler's formula $v - e + f = 2$:

$$2 = v - e + f \leq v - e + \frac{2e}{3} = v - \frac{e}{3},$$

whence $e \leq 3v - 6$. □

Corollary 2.2. *Every simple planar graph G contains a vertex of degree at most 5.*

Proof. Suppose for contradiction that every vertex has degree at least 6. Then the sum of degrees satisfies $\sum_v \deg(v) \geq 6v$. By the handshaking lemma, $2e \geq 6v$, so $e \geq 3v$. But by Corollary 2.1, $e \leq 3v - 6$, giving $3v \leq 3v - 6$, a contradiction. $\square$

Corollary 2.3. *The complete graph K_5 is non-planar.*

Proof. K_5 has $v = 5$ and $e = 10$. By Corollary 2.1, any simple planar graph on 5 vertices satisfies $e \leq 3(5) - 6 = 9 < 10$. Hence K_5 is not planar. $\square$

Corollary 2.4. *If G is a connected simple planar graph with $v \geq 3$ vertices, e edges, and girth at least 4 (i.e. G contains no triangle), then*

$$e \leq 2v - 4.$$

Proof. Since G has girth at least 4, every face has degree at least 4. Thus $\sum \deg(f_i) \geq 4f$, giving $2e \geq 4f$, i.e. $f \leq e/2$. Euler's formula then yields $2 = v - e + f \leq v - e + e/2 = v - e/2$, so $e \leq 2v - 4$. $\square$

2.3. Non-Planarity of $K_{3,3}$

Theorem 2.2. *The complete bipartite graph $K_{3,3}$ is non-planar.*

Proof. $K_{3,3}$ has $v = 6$ vertices and $e = 9$ edges. Being bipartite, $K_{3,3}$ contains no odd cycles; in particular it contains no triangle, so its girth is at least 4. By Corollary 2.4, any planar graph satisfying these conditions must have $e \leq 2(6) - 4 = 8 < 9$. This contradiction shows $K_{3,3}$ is non-planar. $\square$

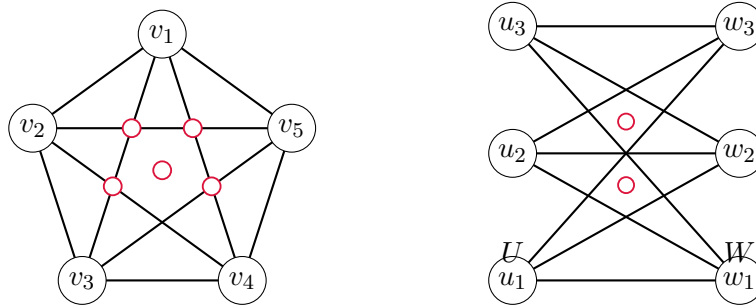


Figure 2: Left: K_5 drawn with five vertices; the circled intersections indicate unavoidable edge crossings, confirming non-planarity. Right: $K_{3,3}$ in its standard bipartite layout; crossings are similarly unavoidable.

§ 3. Kuratowski's Theorem and Planarity Testing

3.1. Subdivisions and Homeomorphic Graphs

Definition 3.1 (Subdivision). A **subdivision** of an edge $uv \in E(G)$ is obtained by replacing uv with a path u, w, v , where w is a new vertex of degree 2. A **subdivision of a graph H** is a graph obtained from H by subdividing any subset of its edges zero or more times.

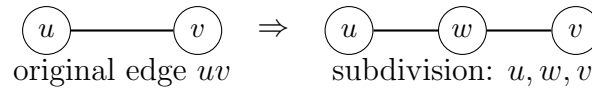


Figure 3: Subdivision of edge uv : the new vertex w has degree 2.

Definition 3.2 (Homeomorphic graphs). Two graphs G and H are **homeomorphic** if there exists a graph F such that both G and H can be obtained from F by a sequence of edge subdivisions. Equivalently, G and H are homeomorphic if and only if they have the same *underlying simple graph* after suppressing all degree-2 vertices.

3.2. Graph Minors

Definition 3.3 (Graph minor). A graph H is a **minor** of G , written $H \preceq G$, if H can be obtained from G by a (possibly empty) sequence of the following operations:

1. **Edge deletion:** remove an edge from the graph.
2. **Vertex deletion:** remove a vertex (and all incident edges).
3. **Edge contraction:** merge the two endpoints of an edge into a single vertex, removing the edge and any resulting multi-edges.

3.3. Kuratowski's and Wagner's Theorems

Theorem 3.1 (Kuratowski, 1930). *A graph G is planar if and only if it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.*

Remark 3.1. The “only if” direction (planar $\Rightarrow$ no forbidden subgraph) follows from Corollaries 2.3 and 2.2 together with the observation that subdivisions preserve planarity. The “if” direction (no forbidden subgraph $\Rightarrow$ planar) is the deep combinatorial content of the theorem; see Diestel [?] for a complete proof.

Theorem 3.2 (Wagner, 1937). *A graph G is planar if and only if it does not contain K_5 or $K_{3,3}$ as a minor.*

Remark 3.2. While Theorem 3.1 provides a complete structural characterisation of planarity, searching exhaustively for homeomorphic subgraphs is computationally expensive. In practice, **linear-time planarity testing algorithms** are used; the landmark result is due to Hopcroft and Tarjan (1974), which decides planarity of an n -vertex graph in $O(n)$ time.

3.4. The Petersen Graph

Example 3.1 (Petersen graph is non-planar). The Petersen graph (Figure 4) is a well-known 3-regular graph on 10 vertices and 15 edges. It is non-planar: one can exhibit a $K_{3,3}$ minor by contracting appropriate edges. By Theorem 3.2, this confirms non-planarity.

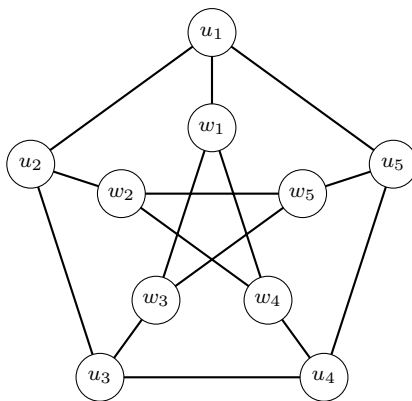


Figure 4: The Petersen graph: 10 vertices, 15 edges, 3-regular. It is non-planar; it contains $K_{3,3}$ as a minor (by contracting edges u_1u_2 , u_3u_4 , and two spokes).

§ 4. Graph Coloring — Vertex Coloring

Definition 4.1 (Proper vertex coloring). A **proper vertex coloring** of a graph $G = (V, E)$ with colors from a set C is a function $\phi: V \rightarrow C$ such that $\phi(u) \neq \phi(v)$ whenever $uv \in E$.

Definition 4.2 (Chromatic number). The **chromatic number** of G , denoted $\chi(G)$, is the minimum cardinality $|C|$ of a color set C admitting a proper vertex coloring of G .

4.1. Standard Examples

Example 4.1. The following chromatic numbers are standard:

- (a) $\chi(K_n) = n$, since every pair of vertices is adjacent.
- (b) $\chi(C_n) = 2$ if n is even; $\chi(C_n) = 3$ if n is odd.
- (c) $\chi(G) = 2$ for any bipartite graph G (with at least one edge).
- (d) $\chi(T) = 2$ for any tree T (with at least one edge), since trees are bipartite.
- (e) $\chi(G) = 1$ if and only if G has no edges (empty graph).

4.2. Bounds on the Chromatic Number

Let $\omega(G)$ denote the **clique number** of G (size of the largest complete subgraph) and $\Delta(G)$ the **maximum degree**.

Proposition 4.1. For any graph G , $\omega(G) \leq \chi(G)$.

Proof. Within a clique of size $\omega(G)$, every vertex is adjacent to every other; thus all $\omega(G)$ vertices must receive distinct colors. $\square$

Proposition 4.2 (Greedy bound). For any graph G , $\chi(G) \leq \Delta(G) + 1$.

Proof. Order the vertices $v_1, \dots, v_n$ arbitrarily. Color v_i with the smallest positive integer not already used by its earlier neighbors. When coloring v_i , at most $\Delta(G)$ earlier neighbors can occupy colors, so color $\Delta(G) + 1$ is always available. $\square$

Theorem 4.1 (Brooks, 1941). *Let G be a connected simple graph. If G is neither a complete graph nor an odd cycle, then $\chi(G) \leq \Delta(G)$.*

Remark 4.1. The proof proceeds by ordering vertices cleverly so that the greedy argument requires only $\Delta(G)$ colors. Specifically, one finds a vertex v such that $G - v$ is connected, then applies a greedy coloring starting from v last. The key case analysis shows that in the non-exceptional cases a color is always saved. A complete proof appears in Diestel [?], Chapter 5.

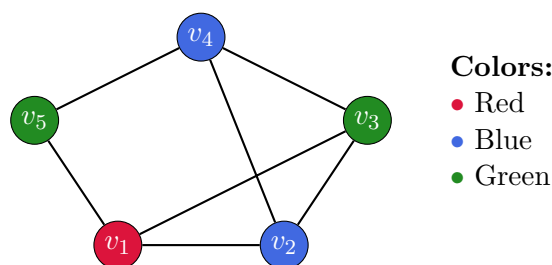


Figure 5: A proper 3-coloring of a 5-vertex graph. No two adjacent vertices share a color. One can verify $\chi(G) = 3$ (the graph contains an odd cycle $v_1v_2v_3v_1$... actually triangle v_1, v_2, v_3 — wait: check edges to confirm $\chi \geq 3$).

§ 5. The Four Color Theorem and Planar Graph Coloring

5.1. Historical Context

The problem of coloring maps so that adjacent regions receive distinct colors was posed by Francis Guthrie in 1852. The conjecture that four colors always suffice remained open for over a century, attracting numerous flawed “proofs”.

Theorem 5.1 (Four Color Theorem — Appel & Haken, 1976). *Every simple planar graph is 4-colorable: for any simple planar graph G ,*

$$\chi(G) \leq 4.$$

Remark 5.1. The proof by Appel and Haken was the first major proof to rely on computer assistance, reducing the problem to a finite case check of 1 936 “unavoidable configurations”. A simplified (though still computer-dependent) proof was given in 1997 by Robertson, Sanders, Seymour, and Thomas. No purely human-verifiable proof is known.

5.2. The Six Color Theorem

We provide a complete proof of the weaker Six Color Theorem, which uses only elementary tools developed thus far.

Theorem 5.2 (Six Color Theorem). *Every simple planar graph G satisfies $\chi(G) \leq 6$.*

Proof. We proceed by induction on $|V(G)| = n$.

Base case. For $n \leq 6$, assign each vertex a distinct color; six colors suffice trivially.

Inductive step. Assume every simple planar graph on fewer than n vertices is 6-colorable. By Corollary 2.2, G contains a vertex v with $\deg(v) \leq 5$. The induced subgraph $G' = G - v$

is a planar graph on $n - 1$ vertices. By the inductive hypothesis, G' admits a proper 6-coloring ϕ .

Now we extend ϕ to v : since $\deg(v) \leq 5$, the vertex v has at most 5 neighbors, which use at most 5 of the 6 available colors. Therefore at least one color remains available for v , and we assign it to v . This yields a proper 6-coloring of G . $\square$

5.3. The Five Color Theorem

Theorem 5.3 (Five Color Theorem — Heawood, 1890). *Every simple planar graph G satisfies $\chi(G) \leq 5$.*

Proof sketch. Proceed by induction on $|V(G)|$. As in the proof of Theorem 5.2, find a vertex v with $\deg(v) \leq 5$ and 5-color $G - v$ by induction. If v has fewer than 5 neighbors, at least one color is free and we extend immediately.

The interesting case is $\deg(v) = 5$, with all 5 neighbors $u_1, \dots, u_5$ (in cyclic order around v) receiving all 5 distinct colors $c_1, \dots, c_5$. Consider the induced subgraph on vertices colored c_1 or c_3 . If u_1 and u_3 lie in *different* connected components of this subgraph, we can swap colors $c_1 \leftrightarrow c_3$ in u_1 's component, freeing color c_1 at u_1 and thus making it available for v . If u_1 and u_3 lie in the *same* component, there is a path P from u_1 to u_3 using only colors c_1 and c_3 . This path together with the edges vu_1, vu_3 forms a closed curve separating u_2 from u_4 and u_5 . Consequently, u_2 and u_4 lie in different components of the (c_2, c_4) -subgraph, and we may swap colors there to free c_2 for v . In either case, v can be colored. $\square$

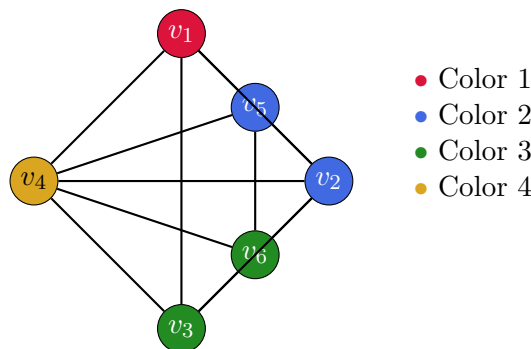


Figure 6: A planar graph with a proper 4-coloring. All four colors are necessary: the subgraph induced by $\{v_1, v_2, v_3, v_4\}$ contains K_4 (the diagonals v_1v_3 and v_2v_4 are present), so at least 4 colors are required.

Remark 5.2. The graph in Figure 6 requires 4 colors because it contains K_4 as a subgraph (vertices v_1, v_2, v_3, v_4 are mutually adjacent via the edges $v_1v_2, v_2v_3, v_3v_4, v_4v_1, v_1v_3, v_2v_4$). By Proposition 4.1, $\chi(G) \geq \omega(G) = 4$, so 3 colors are provably insufficient.

§ 6. Edge Coloring and Chromatic Index

Definition 6.1 (Proper edge coloring). A **proper edge coloring** of a graph $G = (V, E)$ with colors from a set C is a function $\psi: E \rightarrow C$ such that $\psi(e) \neq \psi(f)$ whenever e and f share a common endpoint.

Definition 6.2 (Chromatic index). The **chromatic index** (or **edge-chromatic number**) of G , denoted $\chi'(G)$, is the minimum cardinality of a color set admitting a proper edge coloring of G .

Proposition 6.1. *For any graph G , $\chi'(G) \geq \Delta(G)$.*

Proof. All edges incident to a vertex of maximum degree $\Delta(G)$ must receive distinct colors; hence at least $\Delta(G)$ colors are needed. $\square$

6.1. Vizing's Theorem

Theorem 6.1 (Vizing, 1964). *For any simple graph G ,*

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1.$$

Remark 6.1. The upper bound is the non-trivial content. The proof proceeds by augmenting a partial edge coloring using alternating paths (Kempe chains for edges). A complete proof appears in [?].

Definition 6.3 (Class 1 and Class 2 graphs). A simple graph G is called:

- **Class 1** if $\chi'(G) = \Delta(G)$;
- **Class 2** if $\chi'(G) = \Delta(G) + 1$.

Example 6.1 (Complete graphs). For the complete graph K_n :

- If n is *even*: $\chi'(K_n) = n - 1 = \Delta(K_n)$, so K_n is **Class 1**.
- If n is *odd* ($n \geq 3$): $\chi'(K_n) = n = \Delta(K_n) + 1$, so K_n is **Class 2**.

Example 6.2 (Odd cycles). An odd cycle C_{2k+1} ($k \geq 1$) has $\Delta = 2$. By Vizing's Theorem, $2 \leq \chi'(C_{2k+1}) \leq 3$. Since C_{2k+1} has an odd number of edges and each color class is a matching of even cardinality in a bipartite graph — but C_{2k+1} is not bipartite — one can show that 2 colors do not suffice, so $\chi'(C_{2k+1}) = 3$. Hence odd cycles are **Class 2**.

6.2. König's Line Coloring Theorem

Theorem 6.2 (König, 1916). *For any bipartite graph G , $\chi'(G) = \Delta(G)$. In particular, every bipartite graph is Class 1.*

Proof sketch. The result follows from König's theorem on perfect matchings in bipartite graphs: every regular bipartite graph has a perfect matching. By iteratively decomposing a Δ -regular bipartite supergraph (obtained by adding isolated vertices and a dummy edge to regularize) into perfect matchings, one obtains a proper edge coloring with Δ colors. $\square$

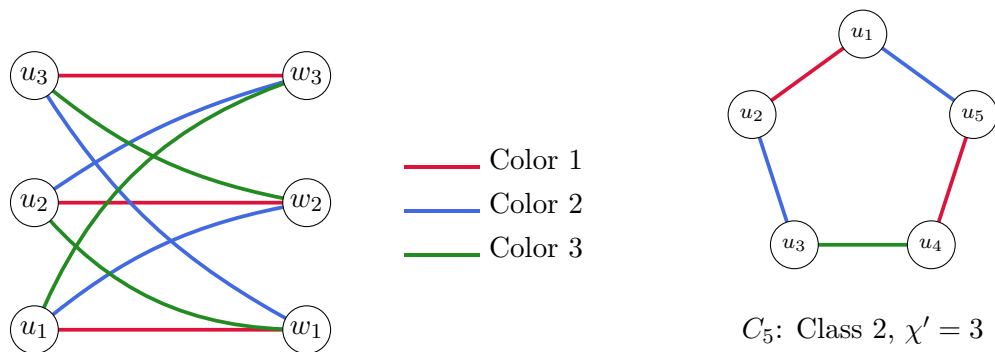


Figure 7: Left: Proper 3-edge-coloring of $K_{3,3}$ ($\Delta = 3$, Class 1 by König's theorem). Each color class is a perfect matching. Right: Proper 3-edge-coloring of C_5 ($\Delta = 2$, $\chi' = 3$, Class 2).